



A four-fuel-stream flamelet model for large-eddy simulation of piloted pulverized coal/ammonia co-combustion

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ABSTRACT

In advanced industrial applications, multiple fuel streams may be introduced in the combustion burners to improve the combustion efficiency and reduce pollutants formation. In particular, the addition of carbon-free sustainable fuel to the existing hydrocarbon burners becomes a feasible and intermediate step to reduce greenhouse gas emissions, e.g. addition of hydrogen to natural gas burners, addition of ammonia to pulverized coal/biomass power plants, etc. To characterize the complex mixing process in such industrial applications, a four-fuel-stream flamelet model is proposed in this work in the framework of large-eddy simulation (LES). The new flamelet model characterizes the mixing among four fuel streams but preserves model consistency for less number of fuel streams. The performance of the proposed flamelet model is first evaluated for a laminar counterflow coal/ammonia co-combustion flame through an *a priori* analysis and then applied to a laboratory-scale piloted turbulent pulverized coal flame with ammonia addition. The simulation results are compared to the detailed chemistry solutions of the counterflow flame and the experimental data of the turbulent flame. The results show that the thermo-chemical quantities, including the major species and radicals, can be accurately predicted for the laminar counterflow flame. For the turbulent piloted flame, the predicted flame structure and the NH, OH, and PAH mass fractions qualitatively agree with the experimental LIF data. The flame temperature and species mass fractions are reasonably predicted as well. Compared with pure coal combustion, the NO formation along the central axis of the burner is suppressed with ammonia addition due to the lower combustibility, caused by the reduced oxygen concentration in the co-combustion. It is, however, increased in the annular regions due to the significant fuel-NO contribution from ammonia combustion. The total NO_x formation in the pulverized coal flame with ammonia addition is slightly decreased compared to the pure pulverized coal flame.

1. Introduction

Compared with hydrogen, ammonia has the advantages of higher boiling point, and easier transportation and storage [1]. However, challenges exist for directly using ammonia as a fuel, such as the low flammability and low radiation. Utilizing ammonia as a co-combustion fuel in existing pulverized coal power plants is an attractive solution to reduce CO₂ emissions.

Both experimental and numerical studies have been conducted to investigate coal/ammonia co-combustion systems. From the experimental side, Tamura et al. [2] investigated ammonia co-combustion in a 1.2 MW thermal pulverized coal furnace with three different ammonia injection methods. Xia et al. [3] reported the fundamental mechanism of the flame propagation phenomenon for pulverized coal/ammonia co-combustion, focusing on the effects of the equivalence ratio of the ammonia/oxidizer mixture on the flame propagating speed in turbulent conditions. Hadi et al. [4] investigated the effect of coal

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type on the turbulent flame speed of coal/ammonia co-combustion under various turbulent intensities. Weng et al. [5] investigated the participation of alkali and sulfur species in solid fuel/ammonia co-combustion applications, focusing on the characteristics of NO emissions and ammonia slip. Tan et al. [6] experimentally investigated the co-combustion characteristics of ammonia with pulverized coal in a drop tube furnace with staged combustion strategy, focusing on the different injection methods and co-combustion ratios. Ma et al. [7] investigated coal/biomass/ammonia co-combustion in a 25-kW furnace, focusing on the NO_x emission, unburnt carbon in fly ash, particulate matter formation, and deposition behaviors.

From the numerical side, Zhang et al. [8] investigated the effect of ammonia co-combustion ratio on the combustion characteristics and NO_x emission of a coal/ammonia co-combustion facility. The eddy-dissipation concept model with the infinitely-fast chemistry assumption was used in the Reynolds-averaged Navier–Stokes (RANS) simulations. Ishihara et al. [9] conducted zero-dimensional numerical calculations for a commercial 100 MW_e pulverized coal-fired boiler with ammonia co-combustion, in which a detailed chemical reaction mechanism is considered using a reactor network. Meller et al. [10] conducted large-eddy simulations (LES) for a laboratory-scale pulverized coal flame with ammonia co-combustion. Based on the assumption that the char combustion is negligible in the pulverized coal burner studied, the mixing among the volatile matter, pilot stream, and the ammonia stream is characterized with a three-mixture fraction flamelet model previously proposed by Wen et al. [11]. Furthermore, a detailed homogeneous kinetics with 129 species and 1644 elementary reactions was introduced, in which both the detailed NH₃ reaction paths and detailed coal chemistry are considered. Yadav et al. [12] conducted LES of coal/ammonia flames with varied ammonia injection locations using the three-mixture fraction flamelet model in [11], in which the mixing among volatile matter, char off-gases, and ammonia is characterized with the respective mixture fractions. While the composition of tar was restricted to the light species C₂H₂ due to limitations of the adopted chemical oxidation mechanism, the sulfur reactions were additionally considered.

For pulverized coal combustion, a flamelet model based on a multi-stream configuration is required due to the co-existence of volatile matter and char off-gases. Multi-stream flamelet models have been developed in previous works for various applications, e.g., the two-stream flamelet model by Hasse and Peters [13] with further extensions by Felsch et al. [14] and Doran et al. [15]; the diluted flamelet/progress variable (FPV) model by Lamouroux et al. [16] for diluted combustion regimes with internal recirculation; the three-stream FPV model by Ihme and See [16] and its applications conducted by Ihme et al. [17] and Chen and Ihme [18]. The two-stream flamelet model was extended to pure pulverized coal combustion by Watanabe and Yamamoto [19].

Although significant progress has been made, the above multi-stream flamelet models cannot describe the mixing in piloted pulverized coal/ammonia co-combustion flames, which includes four fuel streams, i.e., the pilot, volatile matter, char off-gases, and ammonia. In the well-known CRIEPI burner [20], the newly designed experimental setup employs a piloted fuel stream to stabilize the pulverized coal/ammonia flame, which cannot be fully characterized with the existing flamelet models. Note that four fuel streams also exist in coal/ammonia co-combustion with flue gas recirculation (FGR) treatment, piloted coal/biomass co-combustion, etc. Thus, further extension of the flamelet model to four mixture fractions is required to characterize the complex mixing process in the co-combustion combustion systems. In this work, a four-fuel-stream flamelet model is formulated in the framework of LES for simulating the turbulent piloted pulverized coal/ammonia co-combustion flames, which preserves the consistency with the three-mixture fraction flamelet model developed in [11]. The performance of the four-fuel-stream flamelet model is first evaluated on a laminar counterflow pulverized coal/ammonia flame through an *a priori* analysis, and then applied to the CRIEPI burner [20] through fully-coupled flamelet/LES. To the authors' best knowledge, the four-fuel-stream flamelet model is, for the first time, proposed to model piloted coal/ammonia co-combustion.

Table 1

Combustion states corresponding to the limiting values of A , B , C , and Z .

Limits	Combustion states	Limits	Combustion states
$A = 0$	No char off-gases	$B = 0$	No gases from coal
$A = 1$	No volatile matter	$B = 1$	No pilot fuel
$C = 0$	Only ammonia left	$Z = 0$	Only oxidizer left
$C = 1$	No ammonia left	$Z = 1$	No oxidizer left

2. Four-fuel-stream flamelet model

In the proposed four-fuel-stream flamelet model for piloted pulverized coal/ammonia co-combustion, the mixing of the four fuel streams, including the volatile matter, char off-gases, pilot stream, and ammonia, with air is characterized with the corresponding mixture fractions. The mixture fractions of volatile matter Z_{vol} , char off-gases Z_{pro} , pilot stream Z_{pil} , and the ammonia stream Z_{am} are defined as the ratio of the respective mass to the total mass in the local computational cell, i.e.,

$$Z_{vol} = \frac{\xi_{vol}}{\xi_{vol} + \xi_{pro} + \xi_{pil} + \xi_{am} + \xi_{ox}} \quad (1)$$

$$Z_{pro} = \frac{\xi_{pro}}{\xi_{vol} + \xi_{pro} + \xi_{pil} + \xi_{am} + \xi_{ox}} \quad (2)$$

$$Z_{pil} = \frac{\xi_{pil}}{\xi_{vol} + \xi_{pro} + \xi_{pil} + \xi_{am} + \xi_{ox}} \quad (3)$$

$$Z_{am} = \frac{\xi_{am}}{\xi_{vol} + \xi_{pro} + \xi_{pil} + \xi_{am} + \xi_{ox}} \quad (4)$$

where ξ_{vol} , ξ_{pro} , ξ_{pil} , ξ_{am} , and ξ_{ox} are the mass of gas originating from the volatile matter, char off-gases, pilot, ammonia, and the oxidizer streams, respectively. As the values of Z_{vol} , Z_{pro} , Z_{pil} , and Z_{am} depend on each other, the flamelet solutions cannot be mapped to the $(Z_{vol}, Z_{pro}, Z_{pil}, Z_{am})$ space with fixed ranges of trajectory variables. Motivated by the technique proposed by Hasse and Peters [13] for a two-mixture fraction flamelet model, the flamelet solutions are mapped to a unit space through coordinate transformation, $A = Z_{pro} / (Z_{vol} + Z_{pro} + \epsilon)$, $B = (Z_{vol} + Z_{pro}) / (Z_{vol} + Z_{pro} + Z_{pil} + \epsilon)$, $C = (Z_{vol} + Z_{pro} + Z_{pil}) / (Z_{vol} + Z_{pro} + Z_{pil} + Z_{am} + \epsilon)$, and $Z = Z_{vol} + Z_{pro} + Z_{pil} + Z_{am}$, where ϵ is a small positive number, which is set to be 10^{-6} .

Note that the proposed four-fuel-stream flamelet model can be reduced to the three-mixture fraction flamelet model for piloted pulverized coal combustion [11] by setting $Z_{am} = 0$, i.e., $C = 1$. Moreover, it can be further reduced to the conventional two-mixture fraction flamelet model for pure pulverized coal combustion [21] by setting $Z_{am} = Z_{pil} = 0$, i.e., $B = C = 1$. Thus, the developed four-fuel-stream flamelet model preserves model consistency, and the same flamelet table can be applied to different combustion systems firing the same type of coal.

Note also that the proposed flamelet model covers all the limits in the pulverized coal/ammonia co-combustion system, including pure volatile combustion, pure char off-gas combustion, pure pilot combustion, pure ammonia combustion, and the pure oxidizer. The combustion states corresponding to the limiting values of A , B , C , and Z are summarized in Table 1. The trajectory variables A , B , C , and Z are obtained by solving the transport equations for Z_{vol} , Z_{pro} , Z_{pil} , and Z_{am} . In the framework of LES, the Favre-filtered governing equations for these parameters, grouped as $\tilde{Z}_i$, are solved,

$$\frac{\partial \tilde{\rho} \tilde{Z}_i}{\partial t} + \frac{\partial \tilde{\rho} \tilde{u}_j \tilde{Z}_i}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\tilde{\rho} \left(\tilde{\mathcal{G}} + \mathcal{D}_T \right) \frac{\partial \tilde{Z}_i}{\partial x_j} \right] + \tilde{S}_{C, Z_i}, \quad (5)$$

where $i \in \{vol, pro, pil, am\}$, $\bar{\rho}$ is the spatially-filtered gas density, $\tilde{u}_j$ the gas velocity in the j -direction, $\tilde{\mathcal{D}}$ the filtered molecular diffusivity calculated with the unity Lewis number assumption, $\mathcal{D}_T$ the subgrid diffusivity, and $\tilde{S}_{C,Z_i}$ the two-way coupling source term, which is calculated using coal combustion sub-models. In this work, the devolatilization rate is characterized with the Badzioch and Hawksley model [22] and its parameters are the same as in [10]. The char surface reaction is described with the Baum and Street model [23], and the radiative heat transfer is characterized with the P-1 model proposed by Cheng et al. [24]. The other model parameters adopted in the present work follow the suggested values in the CSC/CBCworkshop [25] for the target CRIEPI burner such that a consistent cross-comparison among different research groups worldwide can be ensured. The subgrid viscosity is calculated with the conventional compressible Smagorinsky model [26].

In the flamelet table, the mass transfer between the coal particle phase and gas phase is considered by solving the 1D non-premixed flamelet equations in Z space for different values of A , B , and C . For flamelet modeling of pulverized coal combustion, the flamelet table is generated based on the 1D flamelet equations for gaseous combustion by assuming that the gaseous fuels have been released from the coal particles before reaching the flame front. The heat transfer between the coal particle phase and the gas phase is considered by varying the temperature boundary conditions for the 1D flamelet equations. The fuel composition of species k on the fuel side of the flamelet equations, $Y_{f,k}$, is determined as,

$$Y_{f,k} = \frac{\xi_{vol} Y_{vol,k} + \xi_{pro} Y_{pro,k} + \xi_{pil} Y_{pil,k} + \xi_{am} Y_{am,k}}{\xi_{vol} + \xi_{pro} + \xi_{pil} + \xi_{am}} \quad (6)$$

$$= (1 - A) \cdot B \cdot C \cdot Y_{vol,k} + A \cdot B \cdot C \cdot Y_{pro,k} + (1 - B) \cdot C \cdot Y_{pil,k} + (1 - C) \cdot Y_{am,k},$$

where $Y_{vol,k}$, $Y_{pro,k}$, $Y_{pil,k}$, and $Y_{am,k}$ are the mass fractions of species k in the volatile matter, char off-gases, pilot stream, and the ammonia stream, respectively. The volatile matter is determined using the CRECK-S-C model [27], which describes the detailed devolatilization process based on a multi-step devolatilization model. The resulting species compositions in the volatile matter $Y_{vol,k}$ are [10], $Y_{vol,H_2} = 2.2\%$, $Y_{vol,CO} = 11.9\%$, $Y_{vol,CO_2} = 3.0\%$, $Y_{vol,CH_4} = 0.1\%$, $Y_{vol,NH_3} = 2.7\%$, $Y_{vol,C_5H_5N} = 9.2\%$, $Y_{vol,HCN} = 6.5\%$, $Y_{vol,C_2H_4} = 1.6\%$, $Y_{vol,H_2O} = 3.8\%$, $Y_{vol,C_{10}H_{12}} = 12.0\%$, $Y_{vol,C_6H_6O_2} = 13.5\%$, $Y_{vol,C_6H_5OH} = 10.3\%$, and $Y_{vol,C_{16}H_{10}} = 23.2\%$. The homogeneous kinetics which is used to oxidize this composition is the same as in [10]. The mechanism is developed specifically for coal/ammonia co-combustion by merging the reduced homogeneous oxidation kinetics of coal volatile including large hydrocarbons for tar [28] and the complete ammonia oxidation pathways [29]. Soot formation is not considered in the simulation as the corresponding measurements were not conducted for the coal/ammonia co-combustion burner studied. The fuel composition in char off-gases $Y_{pro,k}$ can be calculated as $Y_{pro,CO} = M_{CO} / (M_{CO} + 0.5\gamma M_{N_2})$ and $Y_{pro,N_2} = 0.5\gamma M_{N_2} / (M_{CO} + 0.5\gamma M_{N_2})$ based on the assumption that char is initially converted to CO in an air atmosphere [21,30]. The species compositions in the pilot $Y_{pil,k}$ and ammonia $Y_{am,k}$ streams are set according to the specific operating conditions.

Considering both mass and heat transfer in the flamelet table, the thermo-chemical quantities, denoted as Ψ , can be parameterized as, $\Psi = \mathcal{F}(A, B, C, Z, Y_{PV}, H_{e,norm})$, where Y_{PV} is the reaction progress variable transformed from the scalar dissipation rate, and is defined as $Y_{PV} = Y_{CO_2} + Y_{H_2O}$. The normalized total enthalpy $H_{e,norm}$ is transformed from the temperature boundaries, and is defined as $H_{e,norm} = (H_e - H_{e,min}) / (H_{e,max} - H_{e,min})$, where H_e is the total enthalpy, and $H_{e,max}$ and $H_{e,min}$ the maximum and minimum total enthalpies, respectively, for specific values of A , B , C , and Z .

In LES, the turbulence-chemistry interactions are considered with the presumed probability density function (PDF). Specifically, a β -PDF

Table 2

The CRIEPI burner experimental conditions [20].

Cases studied	Pure coal	Coal+NH ₃
Coal flow rate [g/h]	465.1	297.4
Air flow rate [l _N /min]	12.0	9.6
NH ₃ flow rate [l _N /min]	0.0	2.4 ^a
H ₂ flow rate [l _N /min]	0.6	0.6

^a NH₃ co-combustion ratio is 20% in lower heating value.

Table 3

The properties of "Coal-5" [20].

Proximate analysis (wt%)		Ultimate analysis (wt%)	
Moisture ^a	0.8	Carbon ^b	70.3
Ash ^b	14.2	Hydrogen ^b	4.57
Volatile matter ^b	32.6	Oxygen ^b	8.78
Fixed carbon ^b	52.4	Nitrogen ^b	1.66
LHV [MJ/kg]	27.9	Sulfur ^b	0.45

LHV: Lower heating value.

^a As received.

^b Dry basis.

is used to characterize the distributions of Z [30,31], and a δ -PDF is employed to describe the subfilter contributions of A , B , C , Y_{PV} , and $H_{e,norm}$ [30]. As the β -PDF of Z depends on its mean, $\tilde{Z}$, and variance, $\tilde{Z}''^2$, the Favre-filtered thermo-chemical quantities in the flamelet look-up table (FLT) can be finally parameterized as,

$$\tilde{\Psi} = \mathcal{G}(A, B, C, \tilde{Z}, \tilde{Z}''^2, \tilde{Y}_{PV}, \tilde{H}_{e,norm}). \quad (7)$$

Except for the transport equations for the mixture fractions in Eq. (5), the Favre-filtered governing equations for $\tilde{Z}''^2$, $\tilde{Y}_{PV}$, and $\tilde{H}_{e,norm}$ are solved according to the formulations in [31].

3. Experimental and numerical setups

The proposed four-fuel-stream flamelet model in the context of LES is applied to a laboratory-scale piloted turbulent pulverized coal/ammonia co-combustion flame experimentally studied at CRIEPI. In this experiment [20,32], pure hydrogen is used as the pilot fuel to stabilize the coal/ammonia co-combustion. The coal particles are transported by the ammonia/air mixture from the inner nozzle with a diameter of 6 mm, and the pilot hydrogen stream is supplied through a 0.5 mm wide annular injector with an inner wall thickness of 0.5 mm and an outer wall thickness of 1 mm. For comparison, the corresponding pure coal combustion case is studied as well, in which the coal particles are transported by a pure air stream with the same momentum input. The experimental conditions and the coal properties are summarized in Tables 2 and 3, respectively. Note that the flamelet table generated for piloted coal/ammonia co-combustion can be directly applied to pure coal combustion with a reduced number of trajectory variables as inputs, which benefits from the consistency of the developed model.

In the numerical setup, a cylindrical domain with a diameter of 50 mm and a total length of 210 mm is used, which has the same mesh resolution as in [31]. The computational domain is discretized with $400 \times 81 \times 56$ grid points in the axial, radial, and azimuthal directions, respectively, resulting in around 2 million cells. The central, pilot, and the coflow inlets are discretized with 15, 4, and 50 grid points, respectively. The shear layer is locally refined to improve spatial accuracy, with the highest mesh resolution of $0.02D$, where D is the central inlet diameter. The overall computational cost for the simulation takes 35,040 CPU-hours (438 h on 80 Intel(R) Xeon(R) Gold 6248 CPU @ 2.50 GHz). The grid sensitivity study has been conducted in a previous work [31] for the same burner. Compared to the finer mesh resolution with 3 million cells, the current mesh resolution with 2

million cells can resolve most of the turbulent kinetic energy and Pope's suggested criterion of 80% resolved turbulent kinetic energy for LES has been satisfied, see Fig. 7 in [31]. The turbulent inlet for the main flow is generated by running a precursor LES of a cyclic pipe with the same diameter of the inner nozzle, and the fully developed flow is used as an unsteady inflow condition. A constant inflow velocity of 1.89 m/s is set for the pilot hydrogen stream, and a small constant velocity of 0.25 m/s is set for the surrounding air flow. For coal/ammonia co-combustion, the species mass fractions in the central inlet are calculated as $Y_{\text{NH}_3} = 0.1295$, $Y_{\text{N}_2} = 0.668$, and $Y_{\text{O}_2} = 0.2025$ according to the designed volume flow rates. The temperatures of the gas flow and the coal particles at the inlet are equal and set to 300 K. The coal particle velocity at the inlet is set to be constant and equals to 7.6 m/s matching the mean velocity of the gas flow at the central inlet. The coal particle diameter lies in the range between 5 μm and 61 μm following a Rosin-Rammler distribution with a mean value of 42.7 μm and a spread parameter of 4.02. At the outlet, the pressure is set to 101325 Pa and the other scalars are subjected to zero-gradient boundary conditions.

Considering that the available experimental data for the CRIEPI burner are limited, the proposed four-fuel-stream flamelet model is also evaluated on a pulverized coal counterflow flame with ammonia addition. The reference data are obtained by conducting detailed chemistry (DC) simulations for the 2D counterflow flame, in which the transport equations are solved for all the species mass fractions in the chemical reaction mechanism as in [11,21]. In the 2D computational setup, coal particles transported by an ammonia/air mixture at the same equivalence ratio as in the CRIEPI burner are injected from the upper port of the counterflow, while an unburnt hydrogen/air mixture at an equivalence ratio of 0.31 flows into the domain from the lower port. The velocities of the coal particles and ammonia/air mixture equal to 1 m/s, while the velocity of the hydrogen/air mixture equals to 2.2 m/s, which enables the flame front staying in the middle of the domain. To evaluate whether the char surface reactions can be handled with the flamelet model, the coal particle diameter is set to be uniform and equals to 5 μm , which promotes surface reactions. The burnt gases and particles leave the domain from the other sides. The other computational setup of the counterflow, such as the domain size, mesh resolution, etc., is referred to [11,21]. The laminar counterflow flame is employed to evaluate the suitability of the trajectory variables and the flamelet table independently. The laminar counterflow flame with detailed chemistry provides a database, which enables comprehensive

verification of the ingredients of a flamelet model in an *a priori* manner. A full flamelet simulation introduces additional uncertainties associated with the transient effects of particle ignition, i.e., the interpolation error between the steady burning line and the pure mixing line, see the discussions for Fig. 19 in [21]. With the verified trajectory variables and flamelet table in the *a priori* analysis, the fully-coupled flamelet/LES is applied *a posteriori* to the turbulent piloted pulverized coal/ammonia co-combustion and compared with the available experimental data.

The gas phase governing equations are solved using the OpenFOAM package. The convection terms in the momentum equations are discretized with a second-order central differencing method, while they are discretized with the total variation diminishing method in the governing equations for scalars. To avoid instabilities, the Euler implicit scheme is used for time integration with a small time step of 2×10^{-6} s. In fact, the reacting CRIEPI jet flame has a limited Reynolds number and its turbulence intensity is not very high, which relaxes the usual requirement on the second order time accuracy for LES. The same treatment was also used in previous LES works [33,34] for the same burner. For LES of the CRIEPI burner, the simulations run for 125 flow-through times based on the bulk flow velocity of the main jet, corresponding to a physical time of 3.5 s. The flow statistics are obtained by time-averaging the instantaneous solutions, starting from the first 50 flow-through times when the steady state is achieved. For the counterflow simulation, the simulation runs for 0.24 s after the flame front has stabilized.

4. Results and discussion

4.1. *A priori* analysis of the counterflow flame

At first, the performance of the four-fuel-stream flamelet model is evaluated for the counterflow flame with ammonia co-combustion through an *a priori* analysis. In the *a priori* analysis, the trajectory variables are calculated from the detailed chemistry (DC) solutions, in which the transport equations for all species mass fractions are solved. The *a priori* evaluation on the pure pulverized coal flame has been conducted in a previous work, see Fig. 6 in [11]. It was found that except for the intermediate species CO and H_2 that are sensitive to the combustion mode [21], the temperature and major species mass fractions are accurately predicted. The comparisons of various thermo-chemical quantities along the central axis of the counterflow pulverized

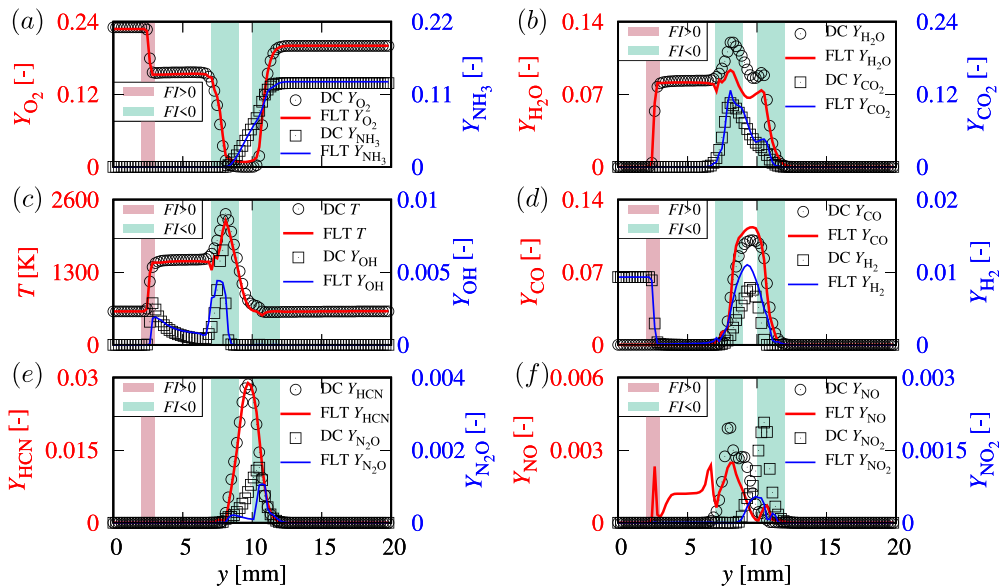


Fig. 1. Comparisons of the thermo-chemical quantities between the detailed chemistry (DC) simulations and the *a priori* flamelet predictions along the central axis of the 2D counterflow. The background color indicates the sign of the flame index ($FI > 0$: premixed combustion mode; $FI < 0$: non-premixed combustion mode).

coal/ammonia flame are shown in Fig. 1. For the major reactants NH_3 and O_2 in Fig. 1(a), the flamelet predictions agree well with the DC solutions. For the major products CO_2 and H_2O in Fig. 1(b), only slight differences of $Y_{\text{H}_2\text{O}}$ exist in certain regions. For the gas temperature and OH radical in Fig. 1(c), good prediction accuracy is obtained, together with the intermediate species CO and H_2 in Fig. 1(d). A local minimum of gas temperature is observed near $y = 7$ mm, at the edge where the non-premixed combustion starts to dominate. This indicates that the thermo-chemical states in the region with the transition of the combustion regime cannot be well represented by the flamelet table generated based on the non-premixed flamelet equations. The intermediate species HCN and N_2O are involved in the prompt and N_2O reaction pathways, and their comparisons are shown in Fig. 1(e). Overall, reasonable predictions are obtained in the whole region. Although good predictions are obtained for the major and intermediate species, large discrepancies exist for the predictions of the NO_x species, see Fig. 1(f). Particularly, local minima and maxima are observed in the profile of the predicted Y_{NO} , which is considered to be related to the existence of the local premixed combustion regime around $y = 2.5$ mm and the transition region between the premixed and non-premixed combustion regimes near $y = 7$ mm. To clarify this, the flame index is introduced, which is defined as $FI = \nabla Y_f \cdot \nabla Y_{\text{O}_2} / |\nabla Y_f \cdot \nabla Y_{\text{O}_2}|$, where $Y_f = Y_{\text{NH}_3} + Y_{\text{H}_2} + Y_{\text{C}_{16}\text{H}_{10}} + Y_{\text{CO}}$. The premixed hydrogen flame is identified as $FI > 0$ at around $y = 3$ mm, where Y_{NO} is over-predicted. This suggests that the prediction accuracy of NO concentration is sensitive to the local combustion mode, which cannot be represented by the present four-fuel-stream flamelet model solely based on non-premixed flamelets. The same finding was obtained for pulverized coal combustion with fuel-bound nitrogen [35].

4.2. Flamelet/LES of the CRIEPI burner

The performance of the four-fuel-stream flamelet model is further evaluated in the LES of the CRIEPI burner. The instantaneous distributions of gas temperature obtained for pure coal combustion and coal/ammonia co-combustion are shown in Figs. 2a and 2b, respectively. The experimental images for the respective cases are superimposed for comparison. Compared with pure coal combustion, the addition of NH_3 leads to a lower combustion intensity in both simulation and experiment. Particularly, the temperature in the central region is substantially lower with ammonia. As indicated by the fraction of volatiles in the particle Y_{vol} in Fig. 2b, the volatile matter has not yet been fully released from the coal particles close to the outlet of the domain with ammonia addition. In fact, for a coal/ammonia co-combustion flame, the coal particle ignition is delayed due to the lower solid-gas ratio. The instantaneous distributions of the various mixture fractions are shown in Figs. 2c and 2d for coal/ammonia co-combustion. Ammonia mixes with the volatiles in the central region, as indicated by Fig. 2c. Char off-gases mix with the pilot stream mainly at around height above burner (HAB) = 30 mm, where char surface reactions start, as shown in Fig. 2d. In the region around HAB = 30 mm, the four fuel streams of volatiles, char off-gases, pilot and ammonia coexist, and the mixing process can only be characterized with the proposed four-fuel-stream flamelet model.

Figs. 3a and 3b compare the predicted OH, NH and PAH mass fractions (right column) with the OH-, NH-, and PAHs-LIFs (left column) for pure coal combustion and coal/ammonia co-combustion, respectively. Here the PAH species is represented by $\text{C}_{16}\text{H}_{10}$, which is the main component of volatile matter. For pure coal combustion in Fig. 3a, the experiment shows two peaks of OH in the radial direction. The inner peak of OH is related to the volatile combustion, while the outer layer is generated by hydrogen combustion in the pilot. Weak PAHs-LIF signals exist close to the inner layer for HAB > 15 mm. These observations of OH and PAHs-LIF signals are reproduced by the simulation. As shown in Fig. 3b, the ammonia addition in the central stream generates an additional inner layer of NH, which already appears close to the burner

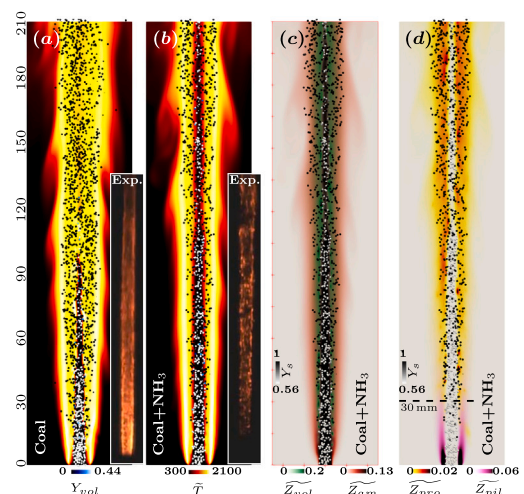


Fig. 2. Instantaneous distributions of gas temperature obtained for (a) pure coal combustion, and (b) coal/ammonia co-combustion. The experimental images for the corresponding cases are shown. Instantaneous distributions of (c) $\tilde{Z}_{\text{vol}}$ and $\tilde{Z}_{\text{am}}$, and (d) $\tilde{Z}_{\text{pro}}$ and $\tilde{Z}_{\text{pil}}$ for coal/ammonia co-combustion. The coal particles in (a, b) are colored by the fraction of volatiles Y_{vol} while those in (c, d) are colored by the solid fraction Y_s . 6% sample particles are superimposed. For clarity of interpretation, an enlarged version of this figure has been added to the Supplementary Material.

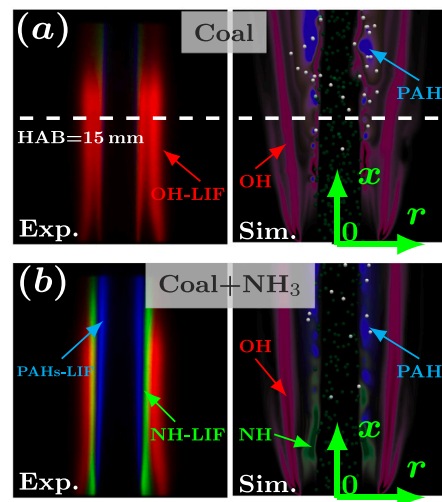


Fig. 3. Comparisons of the OH, NH and PAH mass fractions with the measured OH-, NH-, and PAHs-LIF for (a) pure coal combustion, and (b) coal/ammonia co-combustion. The coal particles colored by the fraction of solid are superimposed (green: $Y_s = 0.56$, white: $Y_s = 1$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

port in both experiment and simulation. This is obviously different from pure coal combustion. With ammonia addition, the combustibility of the fuel mixture decreases due to the lower oxygen concentration, which explains the disappearance of the inner OH layer. The PAH layer exists in the inner layer, which is generated due to the release of volatiles in the shear layer. Although the overall distributions are similar in the streamwise direction, differences in flame spreads can be observed in the radial direction. This could result from neglecting differential diffusion in the flamelet model. Considering that hydrogen is used for the pilot, differential diffusion is expected to affect the species distributions near the burner exit.

Fig. 4 compares the time-averaged temperature $\langle \tilde{T} \rangle$ with the measurement along the central axis of the burner for both pure coal combustion and coal/ammonia co-combustion. The symbol $\langle \cdot \rangle$ indicates

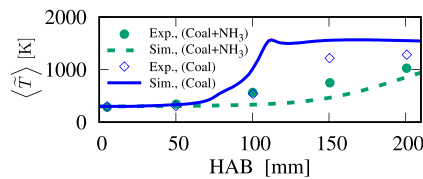


Fig. 4. Comparisons of the predicted temperature with the measurements along the central axis of the burner for both the pure coal and the coal/ammonia co-combustion cases.

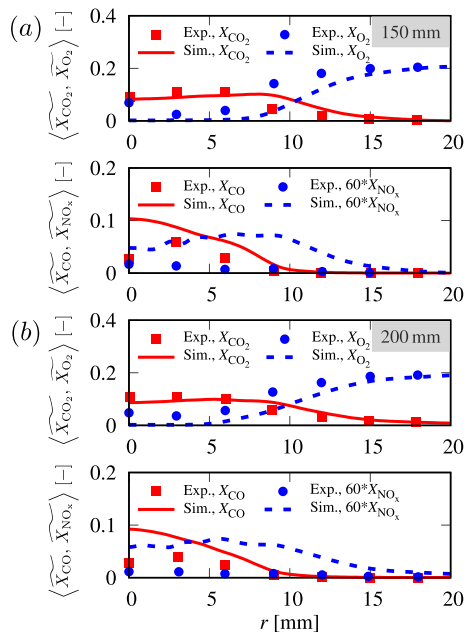


Fig. 5. Comparisons of the mole fractions of CO_2 , O_2 , CO , and NO_x along the radial direction at (a) $\text{HAB} = 150$ mm, and (b) $\text{HAB} = 200$ mm for pure coal combustion.

the time-averaged value. Overall, the temperature is reasonably predicted, particularly for the coal/ammonia co-combustion case. In the downstream, while the temperature increases with increasing HAB for coal/ammonia co-combustion, an almost constant temperature profile is observed for pure coal combustion. These experimental trends are correctly predicted by the proposed four-fuel-stream flamelet model. The mean temperature is over-predicted for the pure coal case in the range $100 \text{ mm} < \text{HAB} < 200 \text{ mm}$, while it is under-predicted for the coal/ammonia co-combustion case. The opposite predictions of the flame temperatures could be related to the transient effects of particle ignition, which cannot be predicted by the steady flamelet model adopted. For the coal/ammonia co-combustion case, the coal particles have not yet been ignited at the central line before leaving the computational domain, see Fig. 2. The previous work by Meller et al. [36] reported that radiation has a negligible effect on the flame temperature prediction for the specific CRIEPI burner studied. Thus, the uncertainties in the prediction are likely not related to radiation modeling.

Next, the species mole fractions in the radial direction are compared with the experiment in Fig. 5 at different HAB. Note that the measurements are available only for pure coal combustion. For the comparisons at $\text{HAB} = 150$ mm, as shown in Fig. 5(a), the values of $\langle \bar{X}_{\text{CO}_2} \rangle$, $\langle \bar{X}_{\text{O}_2} \rangle$, and $\langle \bar{X}_{\text{CO}} \rangle$ are reasonably predicted, while the NO_x concentration calculated as, $X_{\text{NO}_x} = X_{\text{NO}} + X_{\text{NO}_2}$, is over-predicted. The same prediction accuracy is obtained at $\text{HAB} = 200$ mm, as shown in Fig. 5(b). The discrepancies in predictions of the mole fractions of CO and NO_x species shown in Fig. 5 could be related to the existence of the multiple combustion modes since these species are found to be sensitive

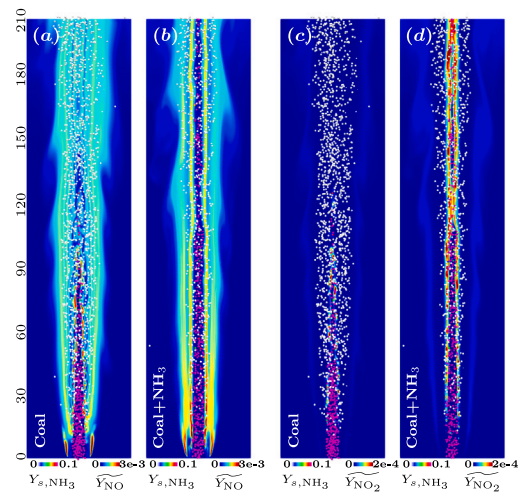


Fig. 6. Instantaneous distributions of the NO and NO_2 mass fractions for (a, c) pure coal combustion, and (b, d) coal/ammonia co-combustion. The coal particles colored by the fraction of volatile- NH_3 in the particle are superimposed. For clarity of interpretation, an enlarged version of this figure has been added to the Supplementary Material.

to the combustion regime [35]. Furthermore, the adopted simplified char reaction model can be also one of the reasons for the discrepancies of CO .

The predictions of the species mole fractions in Fig. 5 can be linked with the temperature predictions in Fig. 4 at the central axis. In the measurement, the CO_2 concentration is significant at $r = 0$ and $\text{HAB} = 150$ mm, which is accurately predicted by the flamelet model, as shown in Fig. 5(a). However, the temperature at the same location is over-predicted, as shown in Fig. 4. The same prediction accuracy is observed at $r = 0$ and $\text{HAB} = 200$ mm. As discussed by Meller et al. [10], the temperature discrepancies might be related to the measurement technique by thermocouples, which is expected to be affected by the turbulent fluctuations.

4.3. NO_x formation characteristics

In this subsection, the NO_x formation mechanism for pure coal combustion and coal/ammonia co-combustion is compared. The instantaneous distributions of the NO and NO_2 mass fractions are shown in Fig. 6 for both cases. The coal particles colored by the fraction of volatile- NH_3 are superimposed. Comparing Figs. 6a and 6b shows that the NO concentration in the shear region is higher in the coal/ammonia case, which is related to the contribution from NH_3 in the carrier stream of the co-combustion case. As indicated by the color of particles, the fraction of volatile- NH_3 in the particle is negligible in the region where the NO concentration is high. Compared to the NO originated from ammonia, the contribution of volatile-N is negligible since the nitrogen element fraction is only 1.66% in the ultimate analysis, see Table 3. However, the NO concentration in the central region of the co-combustion case is negligible, and much smaller than the pure coal combustion case. The decreased NO_x concentration in the central region of the coal/ammonia co-combustion flame was also reported in the experiment [20], and it was attributed to the combined effects of lower combustibility and denitration by high NH_3 concentration. The denitration effects are related to the fact that the denitration pathways $\text{NH}_2 \xrightarrow{+\text{NO}} \text{N}_2$, $\text{NH} \xrightarrow{+\text{NO}} \text{N}_2$, and $\text{N} \xrightarrow{+\text{NO}} \text{N}_2$ are promoted by NH_3 addition [37]. Different from NO , the NO_2 species is mainly located in the central region for both cases, as shown in Figs. 6c and 6d, which is associated with the conversion from NO to NO_2 at low temperature conditions. The NO_2 concentration in the coal/ammonia co-combustion flame is significantly higher than for pure coal combustion due to

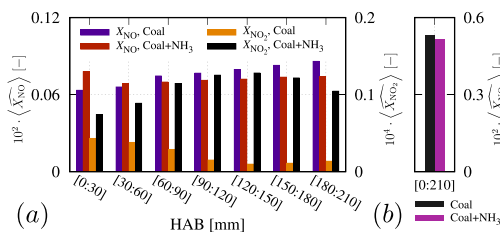


Fig. 7. Time- and volume-averaged values of (a) $\widetilde{X}_{NO}$ and $\widetilde{X}_{NO_2}$ extracted from different sub-domains in the streamwise direction with an equidistant of 30 mm, and (b) the total $\widetilde{X}_{NO_x}$ formation in the whole domain for the coal combustion and coal/ammonia co-combustion cases.

the coexistence of high NO concentration and the low temperature conditions [38].

In coal/ammonia co-combustion, NO_x formation can be either suppressed by the lower oxygen concentration and the denitration effects or promoted by the significant fuel-NO_x pathway. Thus, it is interesting to investigate the net effect. To quantify this, the volume-averaged values of $\widetilde{X}_{NO}$ and $\widetilde{X}_{NO_2}$ are introduced, i.e., $\widehat{X}_k = \int_V \widetilde{X}_k dV / \int_V dV$. Fig. 7a shows the time- and volume-averaged NO_x concentrations in different sub-domains along the streamwise direction. It can be observed that the NO concentration in the coal/ammonia co-combustion flame is slightly higher than in the pure pulverized coal flame for HAB < 60 mm, while it becomes slightly lower in the downstream region where the coal particles are not ignited in the central region. The NO₂ concentration increases with ammonia addition, especially in the downstream region. The total NO_x formation in the entire domain is shown in Fig. 7b. It can be observed that the total NO_x formation for coal/ammonia co-combustion is less than (0.97 times) the one for pure coal combustion, which suggests that ammonia addition suppresses the overall NO_x formation in the computational domain. Note that the comparisons between the two cases are for the early stage of coal/ammonia co-combustion with ammonia slip. Without ammonia slip, NO_x formation in the coal/ammonia co-combustion case is expected to be stronger than for the pure coal combustion case. However, depending on the burner setup and the ammonia injection methods, the NO_x emission can actually stay the same with increasing ammonia co-combustion ratio, e.g., up to 10% [39].

5. Summary and conclusions

In this work, a four-fuel-stream flamelet model in the framework of LES is proposed to simulate piloted turbulent pulverized coal/ammonia co-combustion. The proposed flamelet model is general in terms of the description of the mixing process among four fuel streams, and thus can be applied to different applications with similar mixing process, e.g., coal/ammonia co-combustion with FGR treatment, piloted coal/biomass co-combustion, etc. The performance of the new flamelet model is first evaluated for a laminar counterflow coal/ammonia/hydrogen flame through an *a priori* analysis, and then applied to the laboratory-scale CRIEPI burner through fully-coupled LES. The flamelet tabulated results are compared to the detailed chemistry solutions for the counterflow flame and the available experimental data for the CRIEPI burner. For the counterflow flame, the thermo-chemical variables, including the major species and the radicals involved in NO_x formation, can be accurately predicted with the proposed flamelet model. However, the predictions of NO_x species show discrepancies in regions where the premixed combustion mode is dominant. For the CRIEPI burner, the results show that the overall flame structure, including the experimental images and the NH-, OH-, and PAHs-LIF signals can be captured qualitatively, and the temperature and species mass fractions can also be quantitatively predicted. The overall agreement between the flamelet predictions, and the detailed chemistry solutions

and experimental data confirms the suitability of the proposed four-fuel-stream flamelet model. Comparison of the NO_x concentration in pulverized coal flames with and without ammonia addition shows that the overall NO_x formation is slightly decreased with ammonia co-combustion. This is attributed to the net effects of lower combustibility and denitration, and the significant fuel-NO_x from ammonia combustion.

Novelty and Significance Statement

The novelty of this research is that a four-fuel-stream flamelet model in the framework of large-eddy simulation (LES) is proposed for predicting piloted turbulent pulverized coal/ammonia co-combustion flames. It is significant because:

- The proposed four-fuel-stream flamelet model considers the complex mixing process among volatiles, char off-gases, ammonia and the pilot stream, which can be applied to complicated industrial applications;
- The performance of the proposed four-fuel-stream flamelet model is evaluated on both a laminar counterflow flame through an *a priori* analysis and a laboratory-scale turbulent pulverized coal flame through fullycoupled simulations;
- The overall agreements with the detailed chemistry solutions for the laminar counterflow flame confirm the suitability of the proposed four-fuel-stream flamelet model;
- For the laboratory-scale piloted pulverized coal/ammonia flame, the overall flame structure and temperature are reasonably predicted, and the NOx formation mechanism is analyzed.

CRediT authorship contribution statement

Xu Wen: Performed simulations, Analyzed data, Writing – original draft. **Ali Shamooni:** Contributed to the laminar counterflow simulation, Writing – review & editing. **Oliver T. Stein:** Writing – review & editing, Supervision. **Kazuki Tainaka:** Writing – review & editing, Providing the experimental data. **Dominik Meller:** Writing – review & editing, Providing the chemical mechanism. **Andreas Kronenburg:** Writing – review & editing, Supervision, Funding acquisition. **Andreas M. Kempf:** Writing – review & editing, Supervision. **Christian Hasse:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2024.105470>.

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